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Consistency of dry matter intake in Holstein cows: heritability estimates and associations with feed efficiency

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ABSTRACT

Resilience can be defined as the capacity to maintain performance or bounce back to normal functioning after a perturbation, and studying fluctuations in daily feed intake may be an effective way to identify resilient dairy cows. Our goal was to develop new phenotypes based on daily dry matter intake (DMI) consistency in Holstein cows, estimate genetic parameters and genetic correlations with feed efficiency and milk yield consistency, and evaluate their relationships with production, longevity, health, and reproduction traits. Data consisted of 397,334 daily DMI records of 6,238 lactating Holstein cows collected from 2007 to 2022 at 6 research stations across the United States. Consistency phenotypes were calculated based on the deviations from expected daily DMI for individual cows during their respective feeding trials, which ranged from 27 to 151 d in duration. Expected values were derived from different models, including simple average, quadratic and cubic quantile regression with a 0.5 quantile, and LOESS (Locally Weighted Scatterplot Smoothing) regression with span parameters 0.5 and 0.7. We then calculated the log of variance (**log-Var-DMI**) of daily deviations for each model as the consistency phenotype. Consistency of milk yield was also calculated, as a reference, using the same methods (**log-Var-Milk**). Genetic parameters were estimated using an animal model, including lactation, days in milk and cohort as fixed effects, and animal as random effect. Relationships between log-Var-DMI and traits currently considered in the US national genetic evaluation were evaluated using Spearman's rank

correlations between sires' breeding values. Heritability estimates for log-Var-DMI ranged from 0.11 ± 0.02 to 0.14 ± 0.02 across models. Different methods (simple average, quantile regressions, and LOESS regressions) used to calculate log-Var-DMI yielded very similar results, with genetic correlations ranging from 0.94 to 0.99. Estimated genetic correlations between log-Var-DMI and log-Var-Milk ranged from 0.51 to 0.62. Estimated genetic correlations between log-Var-DMI and feed efficiency ranged from 0.55 to 0.60 with secreted milk energy, 0.59 to 0.63 with metabolic body weight (BW), and 0.26 to 0.31 with residual feed intake. Relationships between log-Var-DMI and the traits in the national genetic evaluation were moderate and positive correlations with milk yield (0.20 to 0.21), moderate and negative correlations with female fertility (-0.07 to -0.20), no significant correlations with health and longevity, and favorable correlations with feed efficiency (-0.23 to -0.25 with Feed Saved and 0.21 to 0.26 with RFI). We concluded that DMI consistency is heritable and may be an indicator of resilience. Cows with lower variation in the difference between actual and expected daily DMI (more consistency) may be more effective in maintaining performance in the face of challenges or perturbations, whereas cows with greater variation in observed versus expected daily DMI (less consistency) are less feed efficient and may be less resilient.

Keywords: genetic correlation, variance, resilience, DMI

INTRODUCTION

Resilience can be defined as the capacity to maintain performance or bounce back to normal functioning after a disturbance (Baggio et al., 2015; Colditz and Hine, 2016; Scheffer et al., 2018; Berghof et al., 2019b). In a livestock production system, disturbances or perturbations may be associated with any factor that can

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cause an animal to be stressed, for example, diseases, social interactions, temperature changes, environmental conditions, management alterations, or diet formulations. These factors are numerous and often unknown, and they can interact with each other, so generalized consistency or resilience may be more valuable than resilience to a specific type of stressor. In dairy cattle, remarkable advances have been made in production efficiency, and as consequence, the environmental impact of dairy farming was reduced (Capper et al., 2009). Still, modern intensive farming systems focus on increasing average performance under optimal conditions, and the ability of an animal to perform in variable or suboptimal conditions has been largely ignored. Part of this can be attributed to the difficulty of obtaining and incorporating environmental data compared with performance data.

The discussion about the potentially negative impact of genetic selection for high production on resilience to management and environmental disturbances is not new (Rauw et al., 1998; Prunier et al., 2010). Selection for resilience is important to ensure dairy farm sustainability (Tendall et al., 2015; Urruty et al., 2016), particularly in the face of climate change. Nevertheless, measuring resilience is not straightforward. Data-driven approaches using high-frequency data can be promising tools for identifying animals that may be more genetically resilient, using phenotypes that are measured over time as deviations of observed performance from the performance that would have been expected under optimal conditions (Scheffer et al., 2018; Berghof et al., 2019b). Based on this principle of generalized resilience, studies have been developed using alternative indicator traits, such as deviations in BW and egg production in laying hens, and deviations in feed intake and feeding behavior in pigs (Berghof et al., 2019a; Putz et al., 2019; Homma et al., 2021; Bedere et al., 2022). In dairy cattle, deviations in daily milk yield from expected lactation curves have been investigated as resilience indicators (Poppe et al., 2020). Statistical approaches used in such studies are diverse, but the purpose is the same, to capture and quantify fluctuations in a performance trait over time.

It should be noted that changes in daily feed intake patterns of an individual may be among the first observable responses when animals are faced with a perturbation (Nguyen-Ba et al., 2020). These changes may occur in response to an unrecorded management event or environmental challenge (Garcia-Baccino et al., 2021). Indeed, variation in daily feed intake has been suggested as indicator trait in genetic selection for improved resilience (Harlizius et al., 2021). Heritability estimates for log-variance and root mean square error of deviations between observed intakes and regressions

of daily intake on age ranged from 0.12 to 0.27 in pigs (Putz et al., 2019; Homma et al., 2021). Despite evidence that consistency of feed intake can be a good indicator of resilience, no studies have been published on feed intake consistency in dairy cattle. Therefore, our goals were to develop new phenotypes for DMI consistency in Holstein cows, estimate genetic parameters of DMI consistency, estimate genetic correlations between DMI consistency with feed efficiency traits (secreted milk energy, metabolic BW, and residual feed intake), estimate genetic correlations between DMI consistency and milk yield consistency, and evaluate the relationships between DMI consistency and US national genetic evaluations for production, health, fertility, and longevity.

MATERIALS AND METHODS

Data

Data consisted of 397,334 daily DMI records of 6,238 lactating Holstein cows (3,625 primiparous and 2,613 multiparous) collected from 2007 to 2022 at 6 research stations: Iowa State University (Ames, IA; 1,339 cows), Michigan State University (East Lansing, MI; 530 cows), University of Florida (Gainesville, FL; 982 cows), University of Wisconsin-Madison (Arlington, WI; 1,992 cows), USDA-Agricultural Research Service Animal Genomics and Improvement Laboratory (Beltsville, MD; 833 cows), and USDA-Agricultural Research Service US Dairy Forage Research Center (Madison, WI; 562 cows). The DMI records were collected between 50 and 200 DIM with an average of 24.7 ± 4.6 kg per day, with a minimum of 0.6 kg, and a maximum of 53.8 kg. These statistics were calculated after removing, within each cow, daily DMI records that exceeded 3.5 SD of the mean, which represented in total 0.43% of the daily DMI records in the original data set. All procedures were approved by the corresponding Institutional Animal Care and Use Committees. Animals were housed in freestall or tiestall facilities, and daily feed intakes were measured via Roughage Intake Control System (Hokofarm Group, Emmeloord, Flevoland, The Netherlands), Calan Broadbent Feeding System (American Calan, Northwood, New Hampshire, USA), GrowSafe System (Vytelle, Lenexa, Kansas, USA), or manual weigh-back of refusals. In addition, records of daily milk yield, milk composition, and BW were available for all the cows. Details of data collection protocols varied by station, but typically DMI and milk yield data were measured daily, milk components were analyzed at least once per week, and BW were measured daily, weekly, or on multiple consecutive days at the beginning, middle, and end of each trial (Tempelman et al., 2015; Cavani et al.,

2023). Data description per research station is provided in Supplemental Table S1.

Consistency traits

Different DMI consistency traits were calculated based on the variation of deviations between observed and expected daily DMI for each cow during the experiment. On average, feed intake was recorded continuously during 64 d per cow, ranging from 27 to 151 d.

The following steps were used to obtain the phenotypes for consistency: (1) we estimated the expected daily DMI for each cow; (2) the deviations of expected and observed daily DMI were obtained; (3) the variance of those deviations was calculated; and finally (4) the variances were logarithm transformed due to normality assumptions of the models used in this study.

The expected values of daily DMI for a given cow across the experiment period were derived using 5 methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter. The simplest model of using average DMI for a given cow across the experimental method was chosen as a baseline, due to the relatively short duration of the feeding trials. Quantile regression is a method that estimates the conditional median or a different quantile. LOESS regression or local regression is a nonparametric approach able to fit nonlinear trends. We used the base functions *rq* and *loess* in R (R core team, 2017).

For quantile regression models, different quantiles were tested (lower and higher than 0.5) and evaluated using the root mean square error and mean absolute error metrics of 3 subsets of cows with low, medium, and high day-to-day variation in DMI. For LOESS regression models, the evaluation of different span parameters was more arbitrary, since the model that minimizes the sum of squared errors is always the one with smallest span parameter, meaning that it basically copies the curve of observed DMI. The objective of this preliminary analysis was to find quantile and span parameters that would achieve a balance between copying short-term trends in the data (too local) and ignoring long-term differences between cows in shape of the lactation curve (too global). For all 3 subsets, the quantile 0.5 provided low values of root mean square error and mean absolute error. We also evaluated the different methods through graphical visualization; see Figures 1 and 2 for examples of the 5 methods used for DMI prediction for cows with average or high variation of DMI within the experimental period, respectively.

Consistency of milk yield was also calculated using the same methods and during the same period used for consistency of DMI, as a reference for comparison. Cohorts with less than 6 animals were excluded from the data set. Descriptive statistics for DMI consistency (**log-Var-DMI**) and for milk yield consistency (**log-Var-Milk**) traits are shown in Table 1.

Feed efficiency traits

The following feed efficiency traits were considered for each cow: secreted milk energy (MilKE), metabolic BW, and residual feed intake (RFI).

As noted earlier, milk samples were obtained at least weekly for determination of milk composition. Secreted milk energy was calculated by week using the following equation (NRC, 2001), and then averaged by cow to obtain the MilKE per individual.

$$\text{MilKE (Mcal)} = (0.0929 \times \text{fat \%} + 0.0563 \times \text{protein \%} + 0.0395 \times \text{lactose \%}) \times \text{milk yield (kg)}.$$

Similarly, BW were recorded manually on at least 3 consecutive days at the beginning, middle, and end of the experimental period in most studies, whereas other studies recorded BW manually on a weekly basis, and others recorded BW daily using walk-over scales. We used linear regression of measured BW on day of experiment to estimate missing BW records. Metabolic BW was calculated as the cow's average BW^{0.75}.

Residual feed intake for each cow was calculated using the following linear mixed model:

$$\text{DMI} = \text{DIM} + \text{Lact} + \text{Cohort} + b_1\text{MilKE} + b_2\text{mBW} + b_3\Delta\text{BW} + e,$$

where DIM represents the fixed effect of days in milk with 9 levels (grouped by 16-d periods: 50–66, 67–83, 84–100, 101–117, 118–134, 135–151, 152–168, 169–185, > 186), Lact represents the fixed effect of lactation number with 4 levels (1, 2, 3, and 4+), Cohort represents the fixed effect of experiment-treatment with 173 levels, MilKE is secreted milk energy with partial regression coefficient b_1 , mBW is metabolic BW with partial regression coefficient b_2 , ΔBW is change in BW (difference of BW at the end and beginning of each experiment period) with partial regression coefficient b_3 , and e is the random residual of the model assumed to follow a normal distribution $e \sim N(0, \mathbf{I}\sigma_e^2)$, representing RFI.

Descriptive statistics for all traits are shown in Table 1. Note that feed efficiency traits had slightly fewer

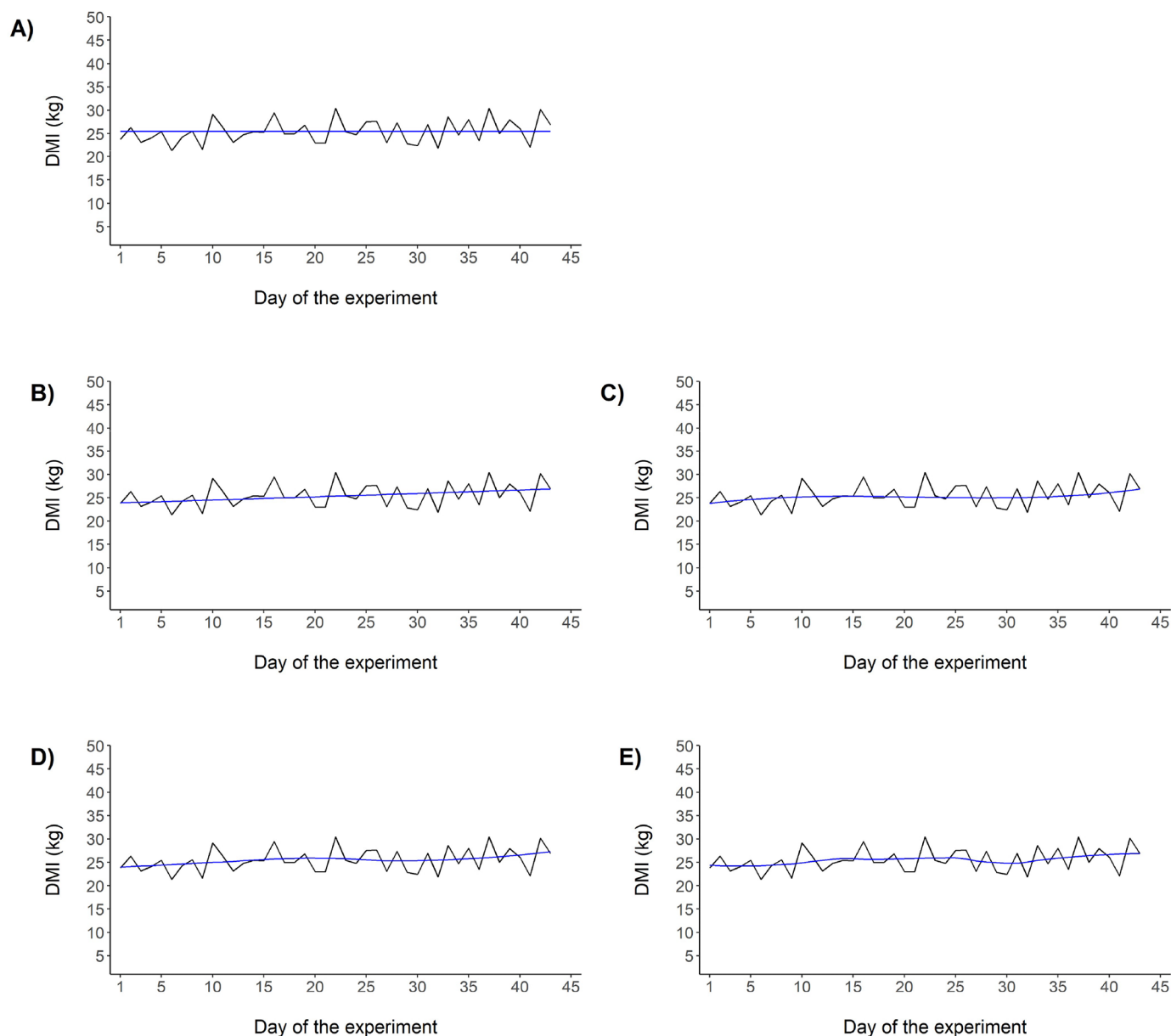


Figure 1. Example of the methods used to estimate the expected daily DMI for a cow with average DMI variation. **A)** cow average (constant), **B)** quadratic quantile regression using a 0.5 quantile, **C)** cubic quantile regression using a 0.5 quantile, **D)** LOESS regression using a 0.7 span parameter, **E)** LOESS regression using a 0.5 span parameter. Black lines show observed DMI, and blue lines show predicted DMI.

records than consistency traits because of some missing BW.

Variance component and breeding value estimation

Estimates of heritability and breeding values for consistency traits, MilkE, and metabolic BW were computed in univariate analyses using the following animal model:

$$y = X\beta + Zu + e,$$

where $\mathbf{y}$ is a vector of phenotypic records log-Var-DMI or log-Var-Milk, β is a vector of fixed effects, $\mathbf{u}$ is a vector of random additive genetic effects, and $\mathbf{e}$ is the vector of random residual effects. Fixed effects included lactation number with 4 levels (1 to 4+), midpoint days in milk with 9 levels (grouped by 16 d), and cohort effect of experiment-treatment with 173 levels. Matrices $\mathbf{X}$ and $\mathbf{Z}$ are incidence matrices relating $\mathbf{y}$ to β and $\mathbf{u}$,

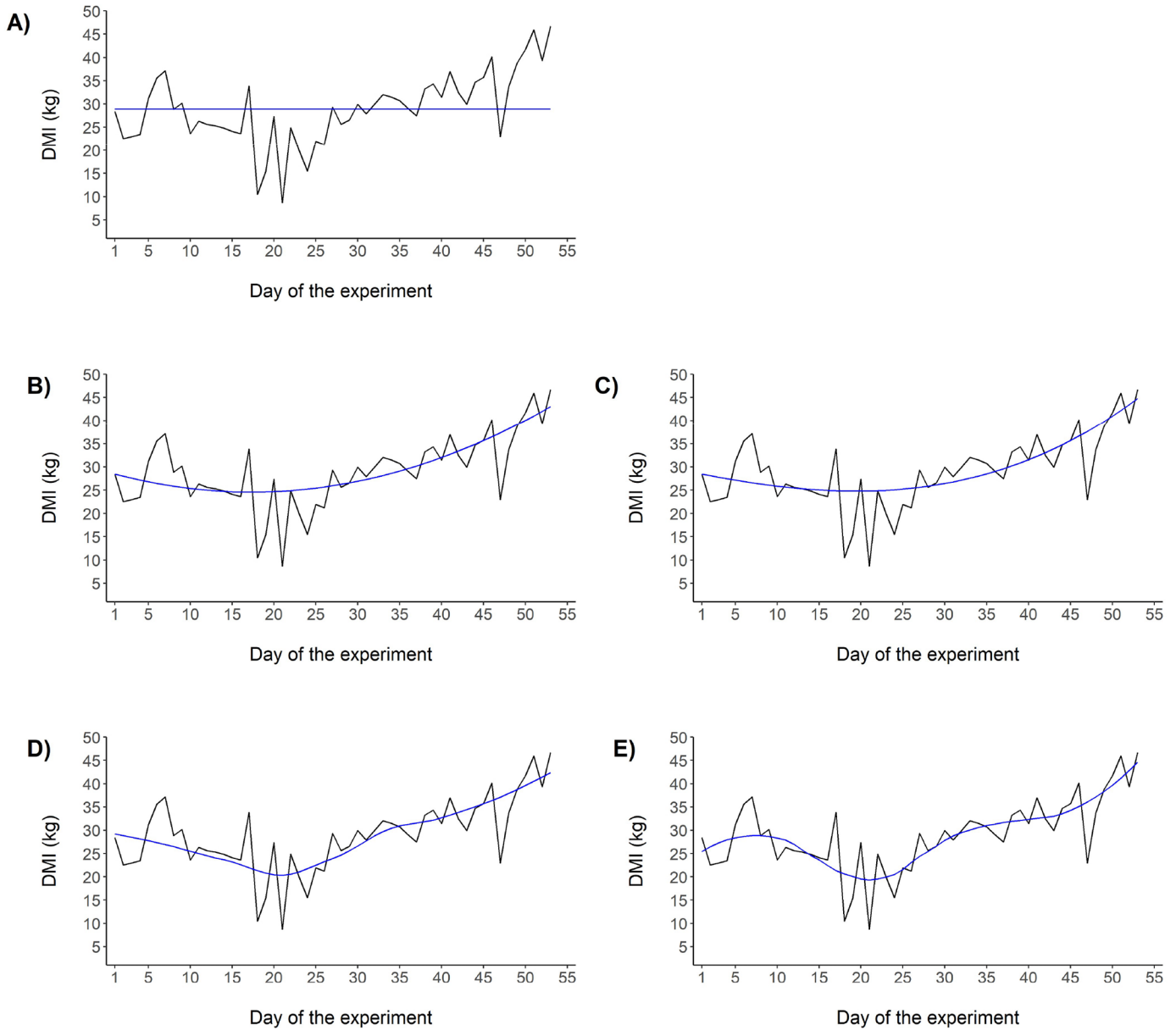


Figure 2. Example of the methods used to estimate the expected daily DMI for a cow with high DMI variation. **A)** cow average (constant), **B)** quadratic quantile regression using a 0.5 quantile, **C)** cubic quantile regression using a 0.5 quantile, **D)** LOESS regression using a 0.7 span parameter, **E)** LOESS regression using a 0.5 span parameter. Black lines show observed DMI, and blue lines show predicted DMI.

respectively. Random effects were assumed to follow a multivariate normal distribution,

$$\begin{bmatrix} u \\ e \end{bmatrix} \sim N \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{A}\sigma_u^2 & 0 \\ 0 & \mathbf{I}\sigma_e^2 \end{bmatrix} \right\},$$

where σ_u^2 and σ_e^2 are the additive genetic and residual variances, respectively; $\mathbf{I}$ is the identity matrix; and $\mathbf{A}$

is the matrix of additive relationships between animals in the pedigree using the last 5 generations.

For RFI, estimates of heritability and breeding values were obtained using the following animal model:

$$y = m + Zu + e,$$

where $\mathbf{y}$ is a vector of RFI records, $\mathbf{u}$ is a vector of random additive genetic effects and $\mathbf{e}$ is the vector of random residual effects. Matrix $\mathbf{Z}$ is an incidence ma-

Table 1. Descriptive statistics for consistency and feed efficiency traits in lactating Holstein cows

Traits ¹	No. of cows	Mean	SD	Minimum	Maximum
<i>Consistency of DMI</i>					
log-Var-DMI					
Average	6,219	1.73	0.63	−0.64	4.17
Quadratic quantile regression	6,219	1.55	0.65	−0.83	3.67
Cubic quantile regression	6,219	1.50	0.65	−0.86	3.72
LOESS with 0.7 span	6,219	1.40	0.66	−0.92	3.54
LOESS with 0.5 span	6,219	1.32	0.67	−0.94	3.51
<i>Consistency of milk yield</i>					
log-Var-Milk					
Average	6,219	2.20	0.78	−0.32	5.29
Quadratic quantile regression	6,219	1.90	0.73	−0.48	4.58
Cubic quantile regression	6,219	1.84	0.73	−0.54	4.40
LOESS with 0.7 span	6,219	1.72	0.72	−0.70	4.21
LOESS with 0.5 span	6,219	1.61	0.72	−1.09	4.15
<i>Feed efficiency traits</i>					
DMI (kg)	6,238	24.7	4.6	0.6	53.8
Secreted milk energy (Mcal)	6,099	29.56	5.46	8.91	49.25
Metabolic body weight (kg^{0.75})	6,099	125.7	11.9	94.9	176.2
RFI (kg)	6,099	0	1.63	−10.39	10.22

¹**log-Var-DMI:** logarithm of the variance of deviations from expected daily DMI; **log-Var-Milk:** logarithm of the variance of deviations from expected daily milk yield; **RFI:** residual feed intake. Expected values of daily DMI and milk were derived using five methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

trix relating $\mathbf{y}$ to $\mathbf{u}$. Random effects were assumed to follow a multivariate normal distribution,

$$\begin{bmatrix} u \\ e \end{bmatrix} \sim N \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{A}\sigma_a^2 & 0 \\ 0 & \mathbf{I}\sigma_e^2 \end{bmatrix} \right\},$$

where σ_a^2 and σ_e^2 are the additive genetic and residual variances, respectively; $\mathbf{I}$ is the identity matrix; and $\mathbf{A}$ is the matrix of additive relationships between animals in the pedigree using the last 5 generations.

Estimates of genetic correlations among DMI consistency traits, between DMI consistency and milk yield consistency, and between DMI consistency and feed efficiency traits were computed in bivariate analyses. Fixed effects were the same used in univariate analyses, and random effects were assumed to follow a multivariate normal distribution,

$$\begin{bmatrix} u \\ e \end{bmatrix} \sim N \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{G}_0\mathbf{A} & 0 \\ 0 & \mathbf{R}_0\mathbf{I} \end{bmatrix} \right\},$$

where $\mathbf{G}_0$ is the additive genetic direct effects (co)variance matrix; $\mathbf{A}$ is the matrix of additive relationships between animals in the pedigree of the last 5 generations; $\mathbf{R}_0$ is the 2×2 residual (co)variance matrix; and $\mathbf{I}$ is an identity matrix with suitable dimensions.

Variance components were estimated using the REML method to estimate heritability and genetic correlations; breeding values were considered the BLUP

derived from the animal models described above, which were computed after algorithm convergence for variance and covariances estimated. These analyses were performed using software from the BLUPF90 family of programs (Aguilar et al., 2018). Additionally, we assessed the relationships between breeding values of the cows' sires using Spearman's rank correlations.

Relationships with other traits

Relationships between DMI consistency traits and the traits currently considered in the US national genetic evaluations of dairy cattle (production, longevity, health, reproduction, and efficiency) were analyzed using Spearman's rank correlations between sires' breeding values. We considered the sires of the cows with DMI records and sires born in 2013 or later. The sires' breeding values were provided by CDCB (Council on Dairy Cattle Breeding) from the official evaluation released in April 2023.

RESULTS

Genetic parameter for consistency of DMI traits

Additive genetic variances, residual variances, and heritability estimates for DMI consistency traits are shown in Table 2. Heritability estimates for log-Var-DMI ranged from 0.11 ± 0.02 to 0.14 ± 0.02 according to the method used to calculate the expected values of daily DMI. The additive genetic variance estimates

Table 2. Variance component estimates and genetic parameters ($\pm$ SE) for consistency traits in lactating Holstein cows

Traits ¹	Additive genetic variance	Residual variance	Heritability
<i>Consistency of DMI</i>			
log-Var-DMI			
Average	0.025 $\pm$ 0.005	0.197 $\pm$ 0.006	0.11 $\pm$ 0.02
Quadratic quantile regression	0.025 $\pm$ 0.005	0.171 $\pm$ 0.005	0.13 $\pm$ 0.02
Cubic quantile regression	0.024 $\pm$ 0.005	0.167 $\pm$ 0.005	0.13 $\pm$ 0.02
LOESS with 0.7 span	0.024 $\pm$ 0.004	0.158 $\pm$ 0.005	0.13 $\pm$ 0.02
LOESS with 0.5 span	0.025 $\pm$ 0.004	0.154 $\pm$ 0.005	0.14 $\pm$ 0.02
<i>Consistency of milk yield</i>			
log-Var-Milk			
Average	0.039 $\pm$ 0.009	0.331 $\pm$ 0.010	0.11 $\pm$ 0.02
Quadratic quantile regression	0.037 $\pm$ 0.008	0.317 $\pm$ 0.009	0.10 $\pm$ 0.02
Cubic quantile regression	0.034 $\pm$ 0.008	0.317 $\pm$ 0.009	0.10 $\pm$ 0.02
LOESS with 0.7 span	0.036 $\pm$ 0.008	0.299 $\pm$ 0.009	0.11 $\pm$ 0.02
LOESS with 0.5 span	0.036 $\pm$ 0.008	0.295 $\pm$ 0.009	0.11 $\pm$ 0.02

¹**log-Var-DMI:** logarithm of the variance of deviations from expected daily DMI; **log-Var-Milk:** logarithm of the variance of deviations from expected daily milk yield; **RFI:** residual feed intake. Expected values of daily DMI were derived using five methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

were similar across methods (0.024 to 0.025). However, the residual variance was higher (0.197) for the average method, resulting in a lower heritability estimate. Although the methods of quadratic quantile regression, cubic quantile regression, and LOESS regression (span parameter of 0.7) resulted in a slight decrease in the residual variance estimates, no significant differences on heritability estimates were observed (0.13 ± 0.02). The lowest residual variance for log-Var-DMI was observed when the LOESS regression model with a span parameter of 0.5 was used to calculate the expected DMI, yielding the highest heritability estimate (0.14 ± 0.02).

Relationships among consistency of DMI traits

The relationships among the log-Var-DMI traits obtained with the 5 different methods were assessed using 2 approaches: genetic correlation estimates among the 5 DMI consistency traits and Spearman's rank correlations of sires' breeding values (Figure 3). Genetic correlation estimates were positive and strong among methods, ranging from 0.94 to 0.99. Standard errors (SE) of genetic correlation estimates were ≤ 0.03 (Supplemental Table S2). Strong and positive Spearman's rank correlations of sires' breeding values were also observed across different traits (i.e., different methods), ranging from 0.92 to 0.99.

Relationship between consistency of DMI and consistency of milk yield

We investigated potential associations between DMI consistency and milk yield consistency during the same recording period. Milk yield consistency was calculated following the same steps and methods used for DMI consistency. Additive genetic variances, residual variances and heritability estimates for milk yield consistency traits are shown in Table 2. Genetic correlation estimates between log-Var-DMI traits and log-Var-Milk traits are shown in Table 3, as well as Spearman's rank correlations of sires' breeding values between these traits. Genetic correlation estimates ranged from 0.51 ± 0.13 to 0.62 ± 0.12 , and Spearman's rank correlations of sires' breeding values ranged from 0.62 to 0.78. The highest genetic correlation estimates were obtained between log-Var-DMI and log-Var-Milk using quantile regression methods (0.62 and 0.78), whereas the average method yielded the lowest genetic and Spearman's rank correlations (0.51 and 0.62).

Relationships between consistency of DMI and feed efficiency traits

We analyzed the relationships between DMI consistency and feed efficiency traits by estimating the genetic correlation between each DMI consistency trait and MilKE, metabolic BW, and RFI. We also estimated the Spearman's rank correlations between sires' breeding values for these traits. Genetic and Spearman's rank correlation estimates are depicted in Figure 4. Standard errors (SE) of genetic correlation estimates ranged from

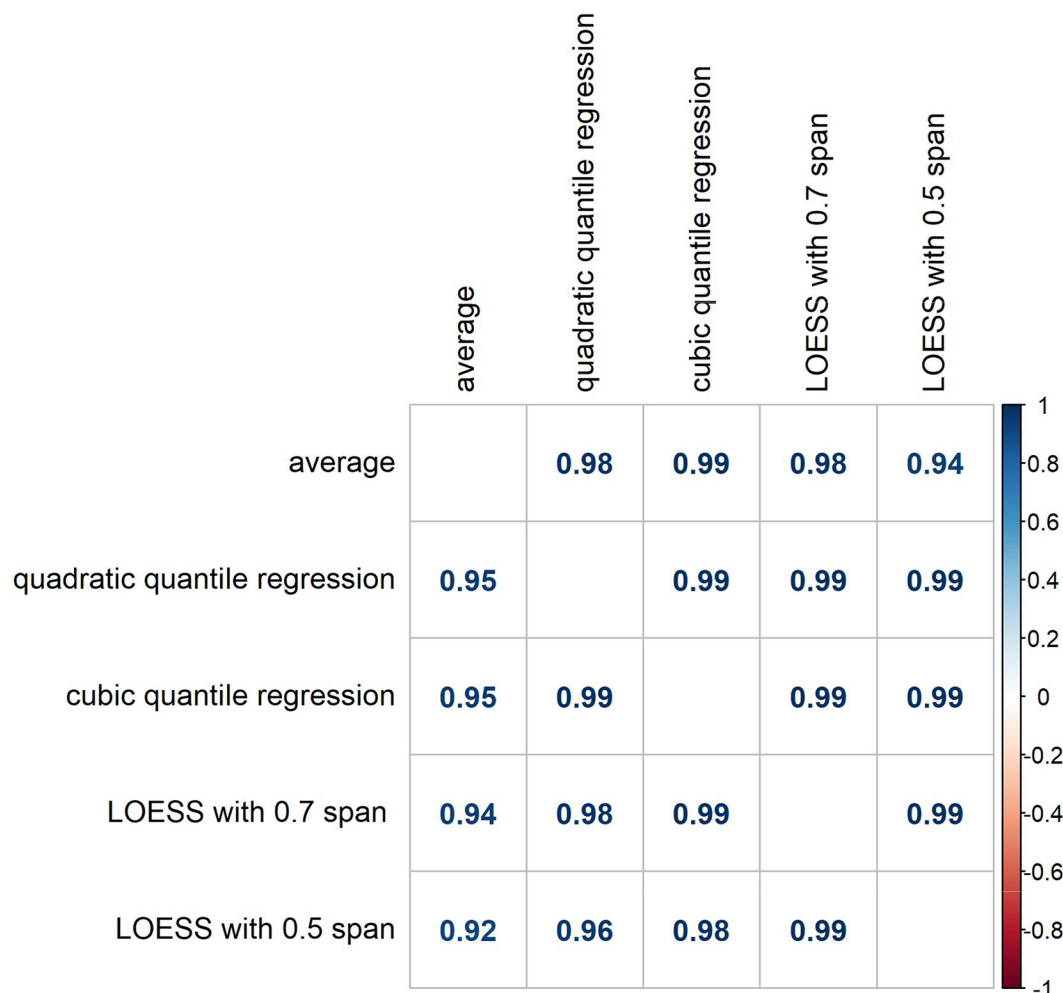


Figure 3. Relationship among DMI consistency traits in lactating Holstein cows: Genetic correlations (above diagonal) and Spearman's rank correlations of 1,214 sires' breeding values (below diagonal). Consistency of DMI was calculated as the logarithm of the variance of deviations from expected daily DMI. Expected values of daily DMI were derived using 5 methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

0.07 to 0.10 (Supplemental Table S1). In general, genetic correlations and sires' breeding value correlations were medium (RFI) to strong (MilKE, metabolic BW) and positive. Genetic and Spearman's correlation estimates between log-Var-DMI traits and MilKE ranged from 0.55 to 0.60 and 0.70 to 0.79, respectively. Genetic and Spearman's correlation estimates between log-Var-DMI traits and metabolic BW ranged from 0.59 to 0.63 and 0.74 to 0.79, respectively. Genetic and Spearman's correlation estimates between log-Var-DMI traits and RFI ranged from 0.26 to 0.31 and 0.29 to 0.36, respectively. The highest genetic correlation estimates and Spearman's rank correlations between log-Var-DMI and RFI were observed using quadratic quantile regression method (0.31 and 0.36), whereas the lowest genetic and Spearman's rank correlations were obtained with

LOESS model with a span parameter of 0.7 (0.26 and 0.29).

Relationships between consistency of DMI and production, longevity, health, reproduction, and efficiency traits

Table 4 shows Spearman's rank correlations of sires' breeding values between DMI consistency traits and 22 traits currently considered in the US national genetic evaluation, grouped as production, longevity, health, reproduction, and feed efficiency traits. The log-Var-DMI traits were weak to moderate and positively correlated with production traits, particularly milk yield (0.20 to 0.21) and protein yield (0.19 to 0.20). In general, no important associations were observed between

Table 3. Relationships between consistency of DMI and consistency of milk yield in lactating Holstein cows analyzed estimating the genetic correlations ($\pm$ SE) and Spearman's rank correlations using 1,214 sires' breeding values

Consistency of DMI traits	Consistency of Milk traits	Genetic correlations	Spearman's correlations of sires' breeding values
log-Var-DMI ¹	log-Var-Milk ¹		
Average	Average	0.51 $\pm$ 0.13	0.62
Quadratic quantile regression	Quadratic quantile regression	0.62 $\pm$ 0.12	0.77
Cubic quantile regression	Cubic quantile regression	0.62 $\pm$ 0.12	0.78
LOESS with 0.7 span	LOESS with 0.7 span	0.59 $\pm$ 0.12	0.78
LOESS with 0.5 span	LOESS with 0.5 span	0.56 $\pm$ 0.12	0.76

¹**log-Var-DMI:** logarithm of the variance of deviations from expected daily DMI; **log-Var-Milk:** logarithm of the variance of deviations from expected daily milk yield. Expected values of daily DMI and milk were derived using five methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

log-Var-DMI traits and longevity and health traits, except gestation length and metritis resistance, which presented correlations from -0.25 to -0.31 and 0.19 to 0.24 , respectively. The log-Var-DMI traits were weak to moderate and negatively correlated with 3 reproduction traits, namely cow conception rate, heifer conception rate, and daughter pregnancy rate, with Spearman's rank correlations ranging from -0.07 to -0.11 , -0.16 to -0.20 , and -0.12 to -0.14 , respectively. Spearman's correlations between log-Var-DMI traits and feed efficiency traits ranged from -0.23 to -0.25 for Feed Saved (efficiency trait in the US national genetic evaluations of dairy cattle), and from 0.21 to 0.26 for RFI.

DISCUSSION

In this study, we assessed different phenotypes for DMI consistency, which can be an indicator of resilience to management and/or environmental perturbations. These phenotypes were calculated based on the variance of deviations between observed and expected daily DMI for individual cows, and different models were used to estimate the expected daily DMI. Due to the inherent natural fluctuations in daily feed intake, it is crucial to consider a model per cow to capture unexpected variations in daily DMI on an individual basis (Tolkamp et al., 2011). Measures of DMI consistency were moderately heritable, suggesting that genetic selection for log-Var-DMI may be successful in achieving more consistent intakes, which may lead to improved feeding management, labor efficiency, and resilience to management and environmental perturbations. Moreover, the genetic correlations among the 5 different methods used to calculate expected daily DMI were close to one, and almost no re-ranking of sires' breeding values across methods was observed. Thus, the log-Var-DMI traits calculated using average,

quadratic quantile regression, cubic quantile regression, or LOESS with varying span parameter did not differ genetically, suggesting that the impact of the different methods used to obtain expected daily DMI is minimal. Nevertheless, curve-fitting methods, such as quantile and LOESS regression, may be more appropriate than a constant (average) value due to lower residual variances. To our knowledge, this is the first genetic study of DMI consistency traits in dairy cattle. However, there are a few studies addressing DMI consistency in pigs. For instance, Homma et al. (2021) reported heritability estimates from 0.12 to 0.27 for log-variance of deviation for daily feed intake using LOESS regression in 3 pig breeds, where the animals were raised in a specific pathogen-free environment with very low mortality rate and health problems. Also in pigs, but under a multifactorial natural disease environment and using crossbred animals, Putz et al. (2019) reported heritability estimate of 0.21 for root mean square error from regressions of daily intake on age. Additionally, the authors explored a different approach, using the proportion of "off-feed" days as a phenotype for individual animals, based on the negative residuals from a quantile regression of feed intake in age. The heritability estimate using this method was 0.15 . The latter approach focuses on identifying a perturbation or estimating the probability of a perturbation in a given period, using data-driven methods and daily feed intake data. Studies using this focus and different statistical methods have been published in pigs and sheep (Nguyen-Ba et al., 2020, Garcia-Baccino et al., 2021).

Interestingly, DMI consistency was associated with milk yield consistency in our study. Genetic correlations between log-Var-DMI and log-Var-Milk traits were strong and positive, indicating that the genetic selection of cows with more consistent daily feed intakes (less day-to-day variation) will also favor more

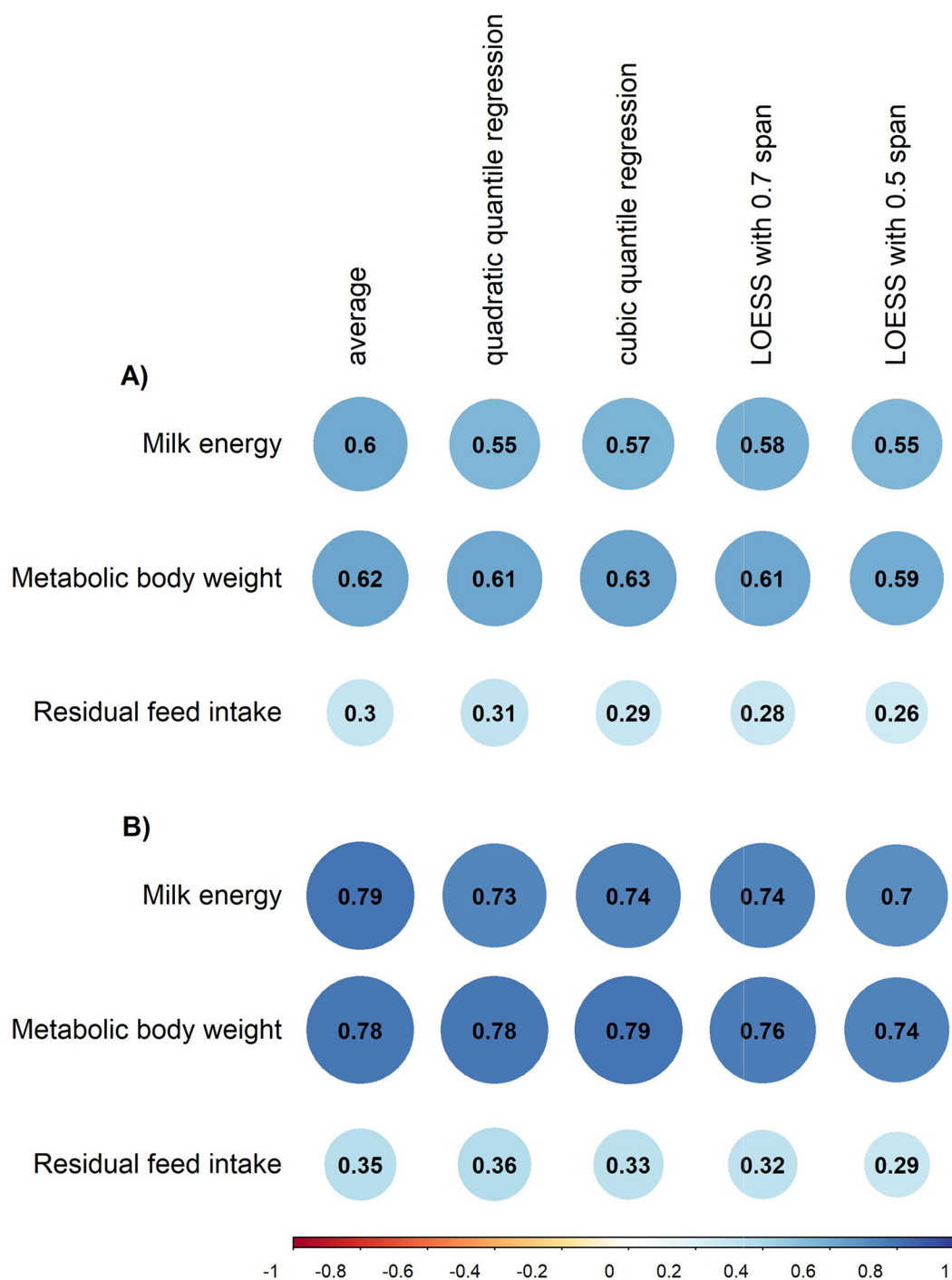


Figure 4. Relationship between DMI consistency traits and feed efficiency traits in lactating Holstein cows: **(A)** Genetic correlations and **(B)** Spearman's rank correlations of 1,199 sires' breeding values. Consistency of DMI was calculated as the logarithm of the variance of deviations from expected daily DMI. Expected values of daily DMI were derived using 5 methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

consistency in milk yield. Recently, a few studies have evaluated daily milk yield consistency over the entire lactation as potential phenotypes for resilience in dairy

cattle. Poppe et al. (2020) computed daily milk yield deviations from lactation curves using different models, and they reported heritability estimates of 0.22 to

Table 4. Relationships between consistency of DMI and traits considered in the US national genetic evaluations of dairy cattle assessed using Spearman's rank correlations of breeding values of 240 sires born in 2013 or later

CDCB genetic evaluation traits ¹	log-Var-DMI traits ²				
	Average	Quadratic quantile regression	Cubic quantile regression	LOESS with 0.7 span	LOESS with 0.5 span
<i>Production</i>					
Milk yield	0.20	0.21	0.21	0.20	0.20
Fat yield	0.14	0.14	0.15	0.16	0.15
Fat %	−0.04	−0.05	−0.04	−0.02	−0.03
Protein yield	0.19	0.20	0.21	0.20	0.19
Protein %	−0.08	−0.08	−0.06	−0.07	−0.06
<i>Longevity</i>					
Productive life	0.09	0.05	0.06	0.08	0.09
Cow livability	0.04	−0.02	−0.01	0.02	0.04
Heifer livability	0.10	0.11	0.13	0.17	0.19
Gestation length	−0.31	−0.26	−0.26	−0.26	−0.25
<i>Health</i>					
Somatic cell score	−0.09	−0.08	−0.09	−0.11	−0.11
Milk fever	0.05	0.04	0.03	0.03	0.01
Displaced abomasum	−0.02	−0.02	−0.04	−0.06	−0.07
Ketosis	0.12	0.10	0.10	0.11	0.12
Mastitis	0.08	0.05	0.08	0.11	0.12
Metritis	0.24	0.20	0.19	0.22	0.23
Retained placenta	0.04	0.03	0.00	0.03	0.03
<i>Reproduction</i>					
Cow conception rate	−0.07	−0.11	−0.11	−0.10	−0.09
Heifer conception rate	−0.18	−0.20	−0.18	−0.16	−0.17
Daughter pregnancy rate	−0.12	−0.13	−0.14	−0.14	−0.13
Early first calving	0.09	0.08	0.07	0.09	0.08
<i>Feed efficiency</i>					
Feed Saved	−0.23	−0.25	−0.24	−0.25	−0.23
RFI	0.21	0.24	0.24	0.26	0.25

¹CDCB: Council on Dairy Cattle Breeding.²**log-Var-DMI:** logarithm of the variance of deviations from expected daily DMI. Expected values of daily DMI were derived using five methods: simple average (constant), quadratic quantile regression using a 0.5 quantile, cubic quantile regression using a 0.5 quantile, LOESS regression using a 0.7 span parameter, and LOESS regression using a 0.5 span parameter.

0.24 for the log-variance of deviations. Elgersma et al. (2018) reported a lower heritability estimate for log-variance of daily milk yield (0.10). Note that in that study the authors used the variance of milk yield observations without considering that expected values may differ according to the lactation curve of each cow. Ben Abdelkrim et al. (2021) discussed a possible association between perturbations in milk yield and BW during the lactation period. It should be noted that no previous studies have shown an association between resilience indicators using both milk yield and feed intake data. Consistency of daily DMI may be more appropriate as a resilience indicator trait compared with consistency of daily milk yield since changes in daily DMI patterns may be observed first, and this DMI change may lead to a change in daily milk yield patterns. Although measuring individual feed intakes is challenging, feed intake records are available and have been routinely collected as part of the national genomic evaluation of feed efficiency traits in dairy cattle.

Our measures of DMI consistency presented a strong relationship with feed efficiency traits. These genetic correlation estimates suggest that cows that are less consistent in daily DMI (higher log-Var-DMI) are less feed efficient (i.e., higher secreted milk energy, greater metabolic BW, and greater RFI). In lactating dairy cows, RFI is widely used to measure feed efficiency, because it is calculated as the difference between actual and predicted feed intake after adjustment for known energy sinks. Therefore, among cows within the same cohort with the same predicted energy needs, those with low RFI eat less than predicted and are considered more efficient (VandeHaar et al., 2016). Our results support that DMI consistency could be an indicator of general resilience. Cows with lower variation in their expected daily DMI may devote their capacity to maintaining performance in the face of challenges or perturbations, thereby demonstrating higher efficiency.

Studies in other species reported similar relationships. For example, phenotypic associations were found in beef cattle, where low RFI steers exhibited signifi-

cantly less day-to-day variation in DMI (Parsons et al., 2020). In different pig breeds, genetic correlations between log-variance of daily feed intake deviations and RFI ranged from 0.66 to 0.71 (Homma et al., 2021). Reinforcing our results, Spearman's rank correlations of sires' breeding values for DMI consistency traits and feed efficiency traits considered in the US national genetic evaluation, namely Feed Saved and RFI, were favorable. Feed Saved represents the amount of feed saved in each lactation due to lower maintenance costs and superior RFI (Gaddis et al., 2021). The scale for publishing Feed Saved differs from that of RFI, such that cows with positive values of Feed Saved are considered more efficient.

Estimated correlations between sires' breeding values for DMI consistency traits and those for production traits were unfavorable, particularly for milk yield and protein yield; however, correlations with reproduction traits were favorable, indicating the potential to improve traits such as heifer conception rate by selection for DMI consistency. The unfavorable relationship between DMI consistency and milk yield was expected because cows with higher DMI, and consequently higher milk production, tend to have greater daily variation in both milk yield and DMI. However, based on our results regarding the relationship between DMI consistency and feed efficiency, those cows were not necessarily more efficient. Positive estimates of genetic correlations between log variance of feed intake and production traits in pigs were reported by Homma et al. (2021). Moreover, the unfavorable relationship between log-variance of daily milk yield deviations and total lactation milk production was noted previously by Poppe et al. (2020).

No significant correlations were found between DMI consistency and health traits, based on published sire breeding values by CDCB. Putz et al. (2019) observed a clear association between consistency and animal health, showing that pigs with smaller daily feed intake variances had lower mortality rates and needed fewer veterinary treatments. In that study, they provided animals with a challenging environment by targeting various viral and bacterial diseases to maintain the disease pressure, thus the mortality rate was high during the experiment period (26%). Also in pigs, Harlizius et al. (2020) reported a genetic correlation of -0.48 between sires' breeding values of feed intake variation and progeny survival to a porcine reproductive and respiratory syndrome virus challenge. One potential explanation for the lack of association between DMI consistency and health traits in our study is that the breeding values for health traits are based on farmer-reported incidence data, and detailed physiological phenotypes, such as blood calcium or β -hydroxybutyrate (BHB), are not available on a national scale.

As we discussed previously, genetic studies involving the development of new phenotypes as an indicator of general resilience in livestock vary in terms of the traits used and statistical models employed. Deviations in daily feed intake and feeding behavior have been used in pigs (Putz et al., 2019; Homma et al., 2021). In dairy cattle, most studies have used daily milk yield deviations (Elgersma et al., 2018; Poppe et al., 2020), although recently Poppe et al. (2022) explored daily step count activity. Moreover, BW deviations and weekly egg production deviations have been used in chickens and layers, respectively (Berghof et al., 2019a; Bedere et al., 2022). Statistical methods used in these studies have varied widely, including different methods to calculate the expected values for a given trait (e.g., simple linear, quantile, and LOESS regressions, cohort average, Wilmink curves) and different methods to compute the deviations that are used as resilience phenotypes (e.g., log-variance, skewness, autocorrelation, root mean square error, proportion of negative residuals). Most studies have referred to these new traits as explicit measures of resilience. In our study, we have proposed the terminology of "consistency" when referring to daily deviations between observed and expected performance observed over a period of time, and in the absence of observed or suspected environmental and management perturbations. Interestingly, we showed that different models used to estimate the expected daily DMI yield very similar results, despite the fact that some curve-fitting methods provided a greater reduction in residual variances. We also provided evidence that DMI consistency, defined as the variance of deviations in observed from expected DMI, may be an indicator trait of resilience in dairy cattle, showing that cows with greater variation in daily DMI (less consistency) are less efficient and may be less resilient. Mechanisms underlying DMI consistency could be related to feeding behavior patterns, which play an important role in explaining dairy cow feed efficiency (Brown et al., 2022; Cavani et al., 2022).

CONCLUSION

In this study, traits for DMI consistency were calculated using different methods based on the variance of deviations between observed and expected daily DMI over time for individual Holsteins cows. Consistency of DMI traits were heritable, suggesting that genetic selection may be successful in achieving more consistent intakes, which may lead to improved feeding management, increased labor efficiency, and greater resilience to management and environmental perturbations. Interestingly, DMI consistency was strongly genetically correlated with milk yield consistency, indicating that

the genetic selection of cows with more consistent daily feed intakes (less day-to-day variation) will also favor more consistent daily milk yield. In addition, measures of DMI consistency were strongly genetically correlated with feed efficiency traits, suggesting that cows that are less consistent in daily DMI are also less feed efficient (higher secreted milk energy, greater metabolic BW, and greater RFI). Overall, our results support the idea that DMI consistency may be an indicator of resilience because cows with lower variation in their expected daily DMI may devote their capacity to maintaining performance in the face of challenges or perturbations, and cows with greater variation in daily DMI (less consistency) are less efficient and may be less resilient. Further research and development is needed to include this phenotype into a routine national genetic evaluation.

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